

FitBuddy – Requirement Analysis

Software Requirement Specification (SRS) & Analysis Phase Document

PROJECT NAME	FitBuddy – AI Fitness Plan Generator	CORE TECH	Google Gemini Models
DOCUMENT PHASE	2. Requirement Analysis	AUTHOR / LEAD	Ravi-teja-777

2.1 Executive Problem Analysis

Personalized fitness coaching is often cost-prohibitive, while static fitness apps provide rigid workout templates that fail to account for unique constraints such as home equipment, past injuries, dietary restrictions, and dynamic schedule shifts. **FitBuddy** solves this by leveraging Gemini AI models to generate adaptive, context-aware workout and nutrition routines dynamically.

2.2 Functional Requirements (FR)

Functional requirements define the specific actions and behavior the FitBuddy application must perform when processing user inputs and communicating with Gemini models.

Req ID	Feature Name	Requirement Description
FR-101	User Profiling Input	System must collect biometric data (age, weight, height, gender), fitness level (beginner, intermediate, advanced), primary goal, and available workout gear.
FR-102	Dietary Constraint Engine	System must accept dietary preferences (vegan, keto, high-protein) and food allergies to filter macro suggestions.
FR-103	Gemini Prompt Construction	System must convert form inputs into optimized JSON-enforced prompts for Gemini model processing.
FR-104	Dynamic Schedule Generator	System must return a structured 7-day workout routine with sets, rep ranges, target muscles, and rest intervals.
FR-105	Plan Export & Save	Users must be able to export their generated workout and nutrition plans as formatted PDFs.

2.3 Non-Functional Requirements (NFR)

NFR-201: Performance & Latency The end-to-end plan generation lifecycle, including Gemini API roundtrips and JSON parsing, must render complete results in under 3.5 seconds.	NFR-202: Reliability & Output Integrity The prompt engineering layer must enforce strict response schemas to ensure 99%+ valid structured outputs without hallucinated formatting.
NFR-203: Safety Guardrails System prompts must explicitly instruct Gemini to flag unsafe caloric targets or extreme intensity combinations that violate general health guidelines.	NFR-204: Accessibility & UX The input wizard must be intuitive and accessible across desktop and mobile devices with zero clutter.

2.4 Hardware & Software Requirements

Environment requirements for running, developing, and serving the FitBuddy AI fitness generator engine.

Software Specifications

- **Language Environment:** Python 3.10+ or Node.js v18+
- **AI Framework:** Google Generative AI SDK (Gemini API)
- **Web Frontend:** Responsive HTML5 / React / Tailwind CSS
- **Document Engine:** HTML-to-PDF print engine for export

Hardware & API Key Setup

- **Development Machine:** Dual-Core CPU, 8GB RAM minimum
- **Network:** Broadband internet for cloud API connectivity
- **API Credentials:** Valid Google Gemini API Key configured in .env file

2.5 Use Case Analysis & User Personas

To ensure high usability, requirements were modeled around key target personas utilizing the system.

User Persona	Primary Need	FitBuddy Gemini Solution
Home Workout Enthusiast	Limited equipment (only dumbbells & resistance bands).	Gemini filters exercise database to generate bodyweight and dumbbell-only routines.
Busy Professional	Maximum 30 minutes available daily for exercise.	Generates high-intensity, condensed 30-minute circuit workouts.
Macro-focused Athlete	Needs strict daily protein/carbing ratios matching goals.	Calculates target total daily energy expenditure (TDEE) and outputs exact daily macronutrient split.

2.6 Feasibility Analysis

- **Technical Feasibility:** Google Gemini models provide rapid, highly capable reasoning to synthesize tailored schedules from text inputs without requiring dedicated model training.
- **Economic Feasibility:** Utilizing Gemini API endpoints dramatically lowers infrastructure costs compared to building custom machine learning pipelines.
- **Operational Feasibility:** The user input flow requires no technical knowledge from end-users, ensuring seamless adoption.

Phase Conclusion: The requirement analysis confirms that integrating Google Gemini API into the FitBuddy workflow provides a scalable, low-latency, and flexible platform for personalized automated fitness planning.

